Reg. No. | | | | | | | | | |

Question Paper Code: 13442

B.E. / B.TECH DEGREE EXAMINATION, NOVEMBER / DECEMBER 2017

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech – Information Technology

15CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2015)

Duration: 3 Hours Max.: 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Find the minterm expansion of f(a,b,c,d)= a'(b'+d)+acd'.
2. Perform 2's complement subtraction of 010110 – 100101.
3. Write down the truth table of a full subtractor.
4. Show that when 2 input multiplexers drive another 2 input MUX, the result is a 4 input MUX.
5. Distinguish between ROM and PLA.
6. Write the features of dot matrix display system.
7. When a shift register becomes universal shift register?
8. Differentiate ripple counter from synchronous counter.
9. What is the most important consideration in making state assignment for asynchronous sequential network?
10. What is Static 1 Hazard?

PART B – (5 × 16 = 80 Marks)

11. a) i. Minimize the following function using k-map.

$F(A,B,C,D)=\Pi(0,3,4,5,6,7,10,14,15)$ (8 Marks)

11. a) ii. Simplify the following switching function using k-map

$F(A,B,C,D)=\sum(0,5,7,8,9,10,11,14,15)+\Phi(1,4,13)$. (8 Marks)

OR

11. b) Find a minimum sum of products expression for the following function using Quine-Mc Cluskey method

$f(A,B,C,D,E)= \sum(0,2,3,5,7,9,11,13,14,16,18,24,26,28,30)$ (16 Marks)

12. a) i. Design a code converter to convert BCD 8421 code to Gray code. (8 Marks)

12. a) ii. A system of logic is to be designed which has two outputs and three inputs. One output will be TRUE if an odd number of inputs are TRUE. The other output will be TRUE if only one input is alone TRUE. Draw the truth table, write the corresponding Boolean equations and draw the logic circuit. (8 Marks)

OR

12. b) i. Implement the following Boolean function using 8X1 MUX.

$F(w,x,y,z) = \Sigma m(0,1,2,4,6,9,12,14)$ (8 Marks)

12. b) ii. Construct a Full adder and Full subtractor, and write a HDL program module for the same. (8 Marks)

13. a) i. Design a Binary to Excess 3 code convertor using ROM. (10 Marks)

13. a) ii. Explain the process of parity generation & checking. (6 Marks)

OR

13. b) i. What is PAL? Show how a PAL is programmed for the following logic function X= AB'C + A'BC'+A'B'+AC (10 Marks)

13. b) ii. Discuss about multiplexed display system. (6 Marks)

14. a) Design a parallel/synchronous counter with the following sequence. Use JK flip flops. 0, 1, 3, 5, 7, 0… (16 Marks)

OR

14. b) i. A sequential circuit with two D flip-flops A and B, one input x , and one output z is specified by the following next state and output equations:

A(t+1) = A'+B,B(t+1)=B'x, z =A+B'

A) Draw the logic diagram of the circuit.
B) Derive the state table.
C) Draw the state diagram of the circuit. (12 Marks)

14. b) ii. Implement JK using D flip-flop. (4 Marks)

15. a) Design an asynchronous sequential circuit with two inputs X and Y and with one output Z. Whenever Y is 1, input X is transferred to Z. When Y is 0, the output does not change for any change in X. Use SR latches. (16 Marks)

OR

15. b) i. An asynchronous sequential circuit is described by the following excitation and output function

$$Y = X_1X_2 + (X_1+X_2)Y$$

$$Z = Y$$

A) Draw the logic diagram of the circuit.

B) Derive the transition table, flow table and output map. (8 Marks)

15. b) ii. Give hazard free realization for the following Boolean function $F(a,b,c,d)= \Sigma m(1,3,6,7,13,15)$. (8 Marks)